

Alpha-Beta

```
MinVal(state, alpha, beta){  
    if (terminal(state))  
        return utility(state);  
    for (s in children(state)){  
        child = MaxVal(s,alpha,beta);  
        beta = min(beta,child);  
        if (alpha>=beta) return child;  
    }  
    return best child (min); }
```

alpha = the **highest** value for **MAX** along the path

beta = the **lowest** value for **MIN** along the path

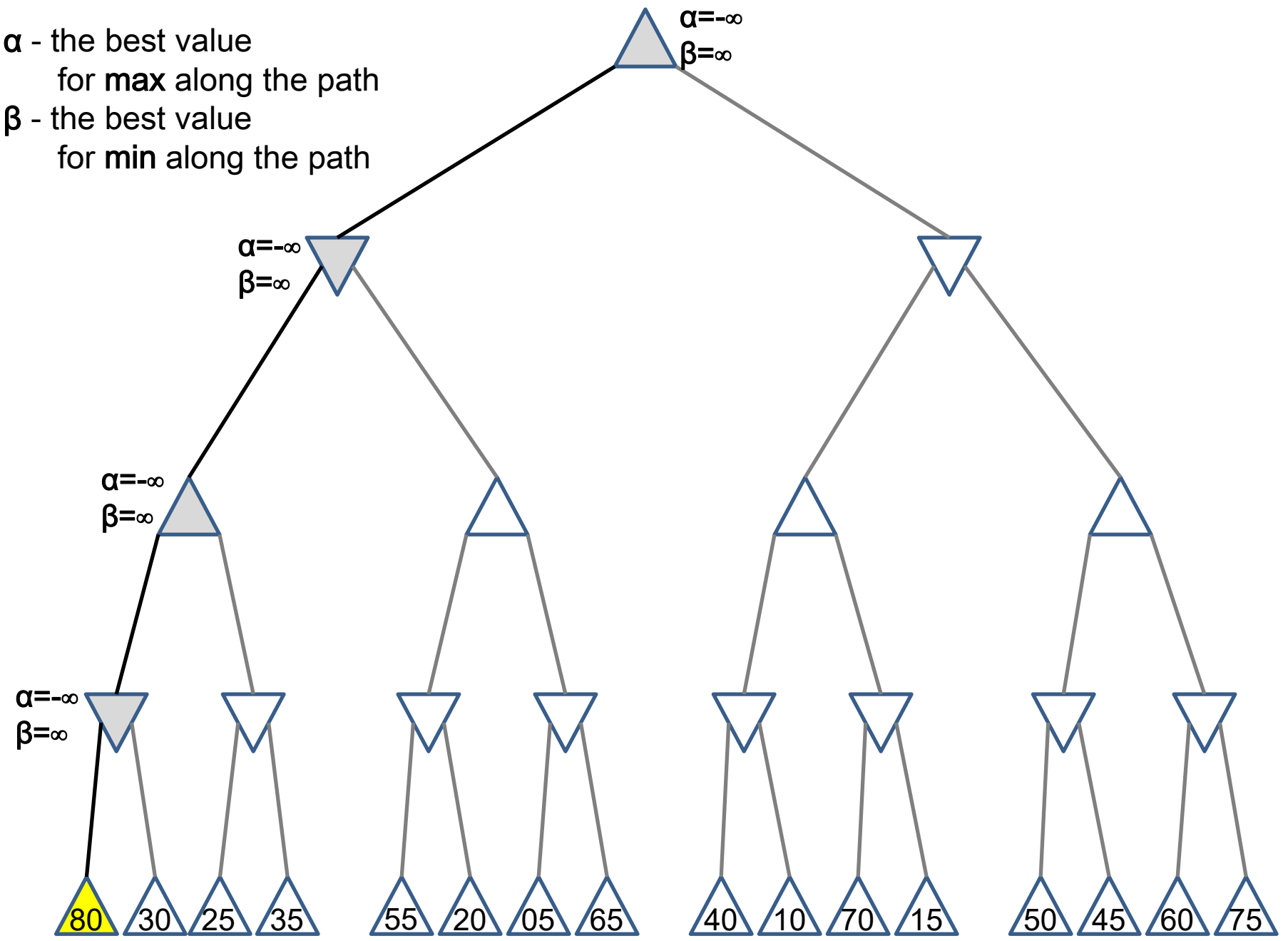
Alpha-Beta

```
MaxVal (state, alpha, beta) {  
    if (terminal(state))  
        return utility(state);  
    for (s in children(state)) {  
        child = MinVal(s, alpha, beta);  
        alpha = max(alpha, child);  
        if (alpha >= beta) return child;  
    }  
    return best child (max); }
```

alpha = the **highest** value for **MAX** along the path

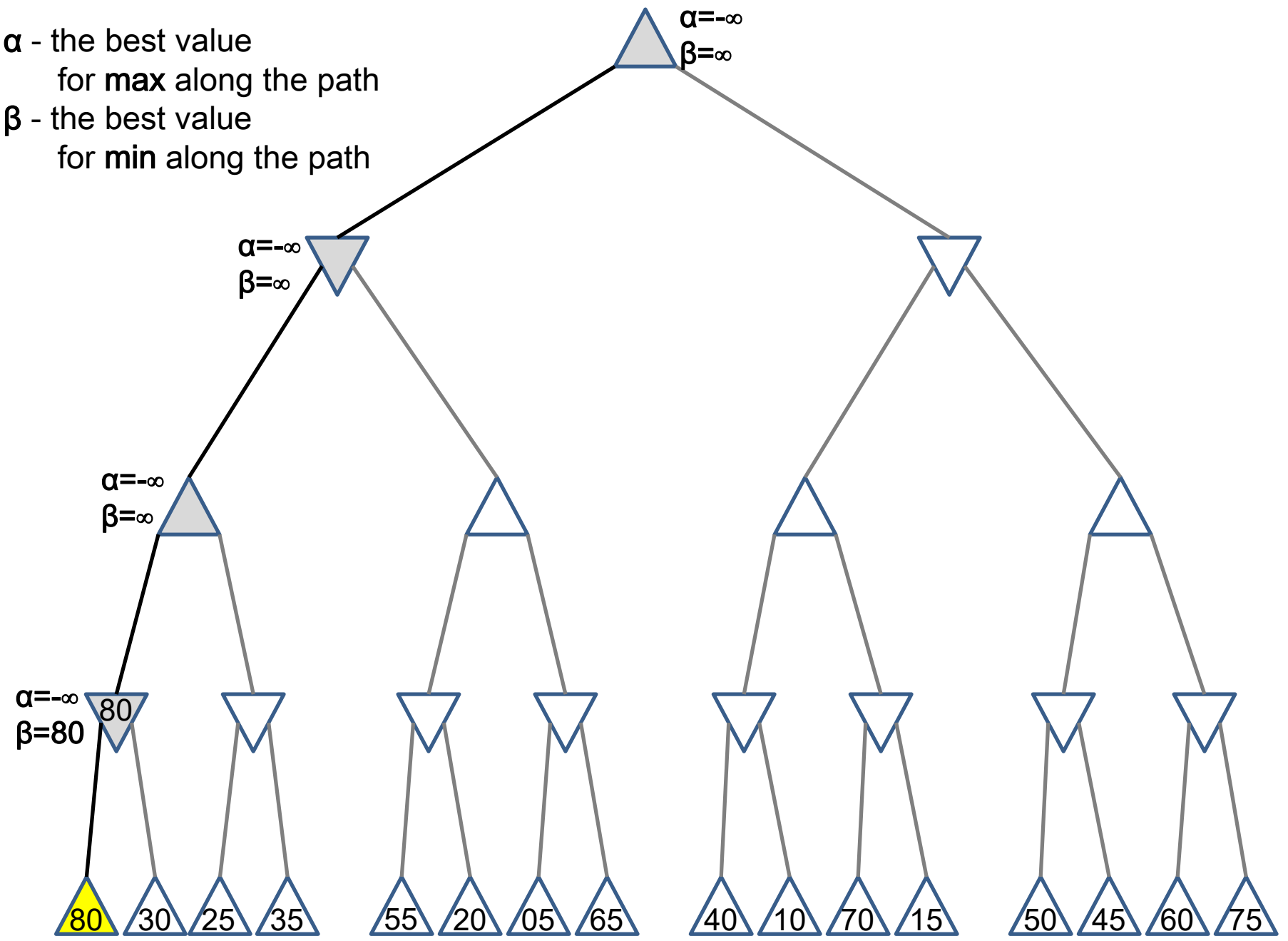
beta = the **lowest** value for **MIN** along the path

α - the best value
for **max** along the path
 β - the best value
for **min** along the path



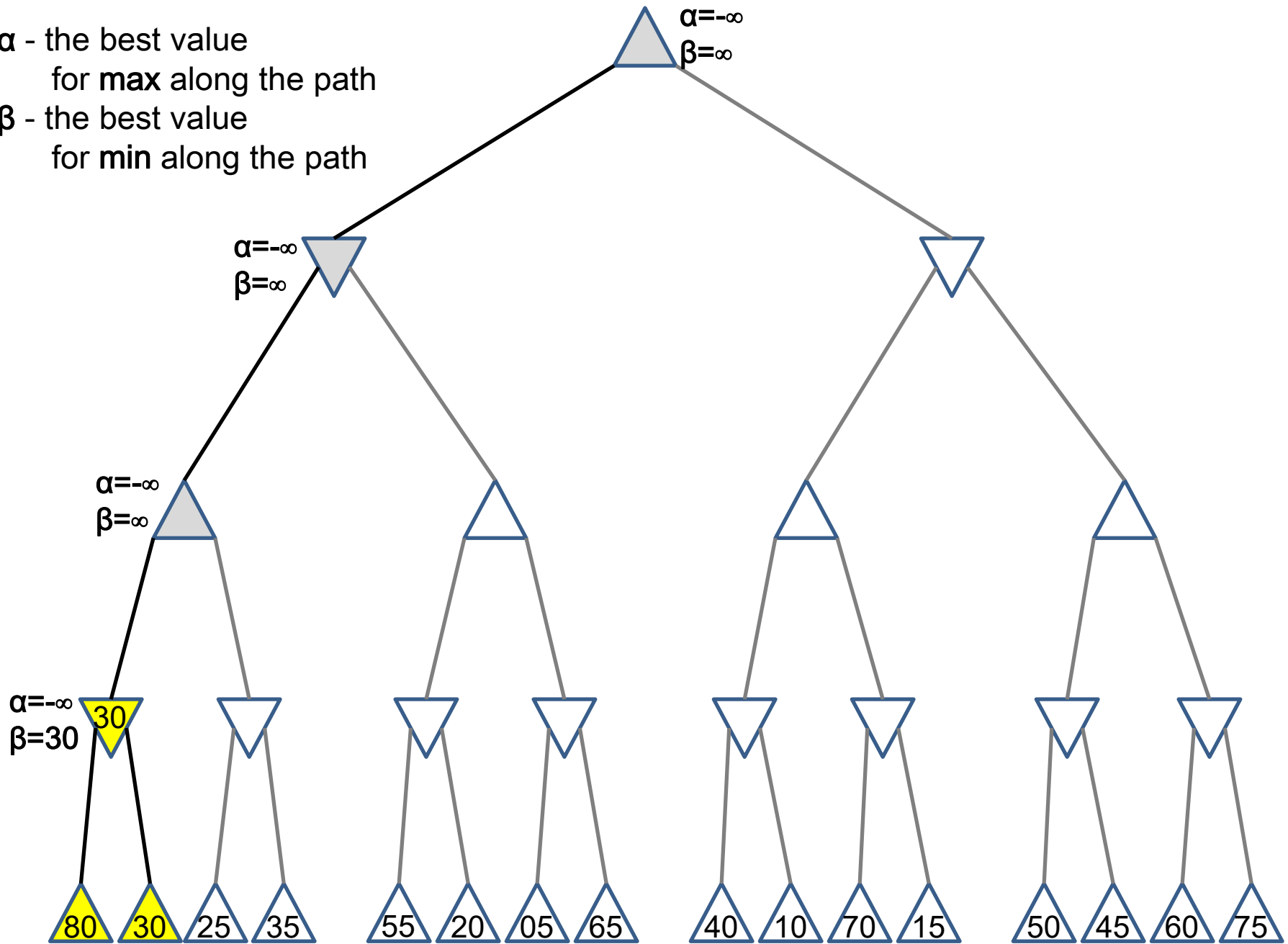
α - the best value
for **max** along the path

β - the best value
for **min** along the path

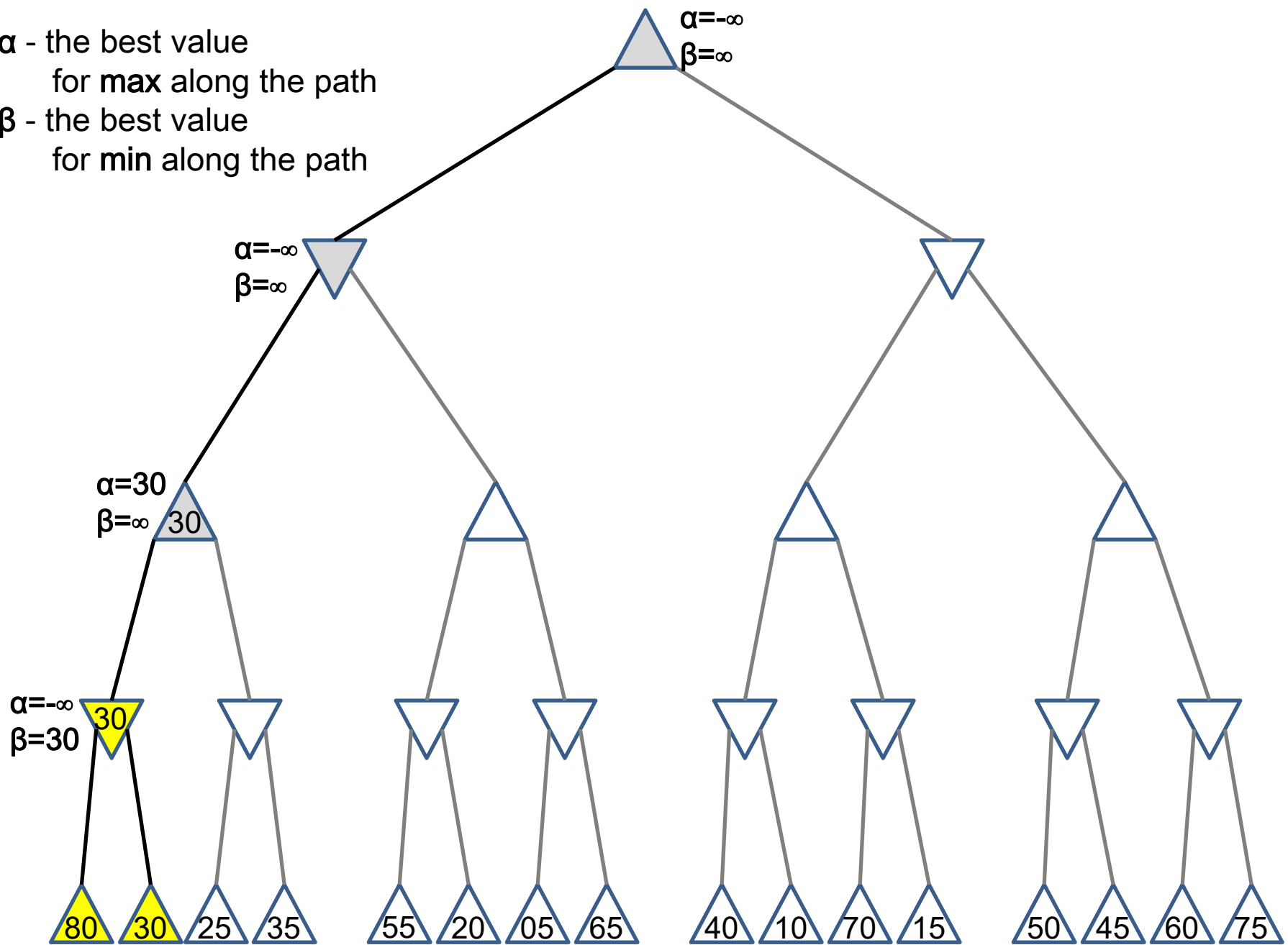


α - the best value
for **max** along the path

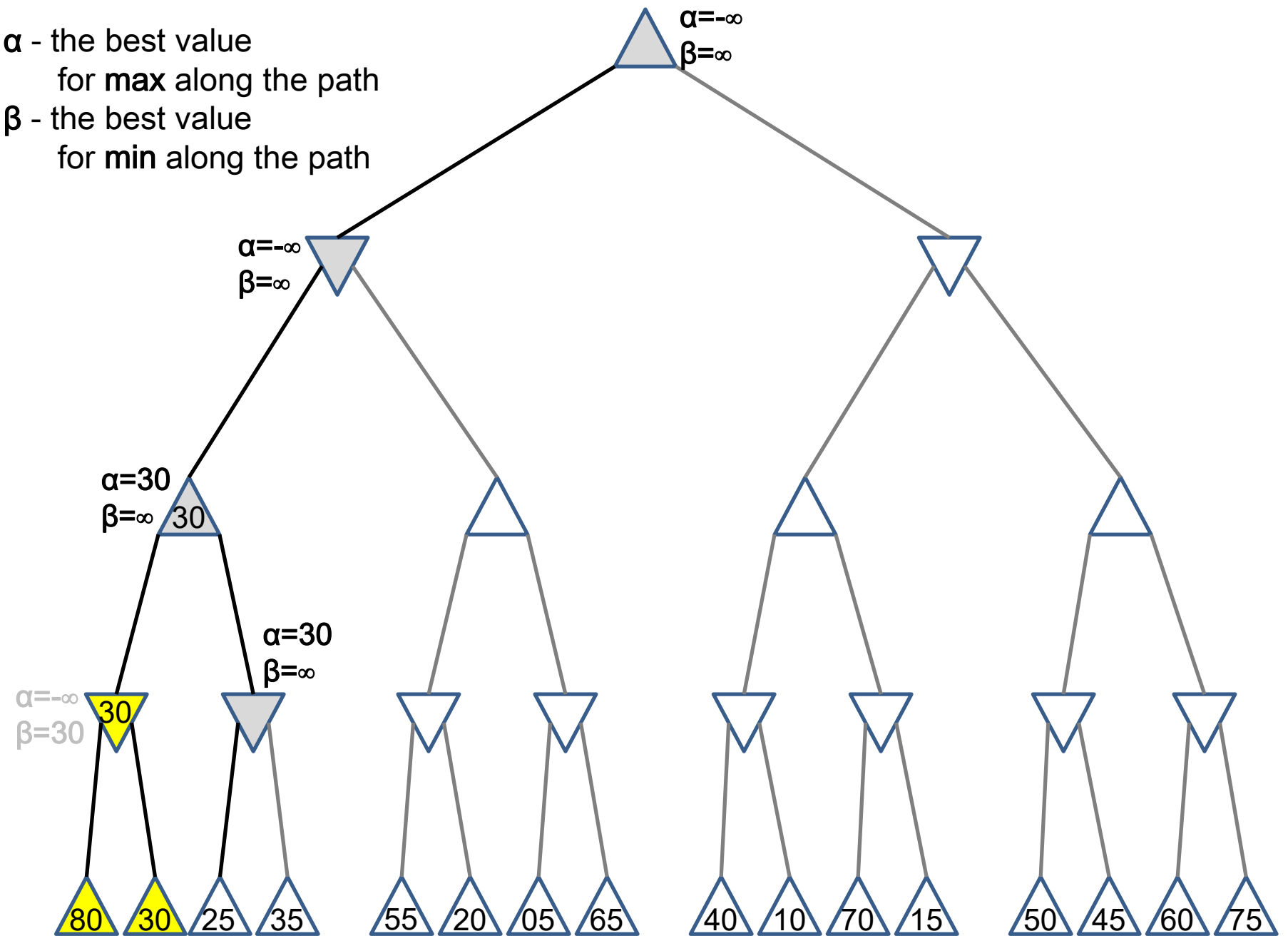
β - the best value
for **min** along the path



- α - the best value for **max** along the path
- β - the best value for **min** along the path

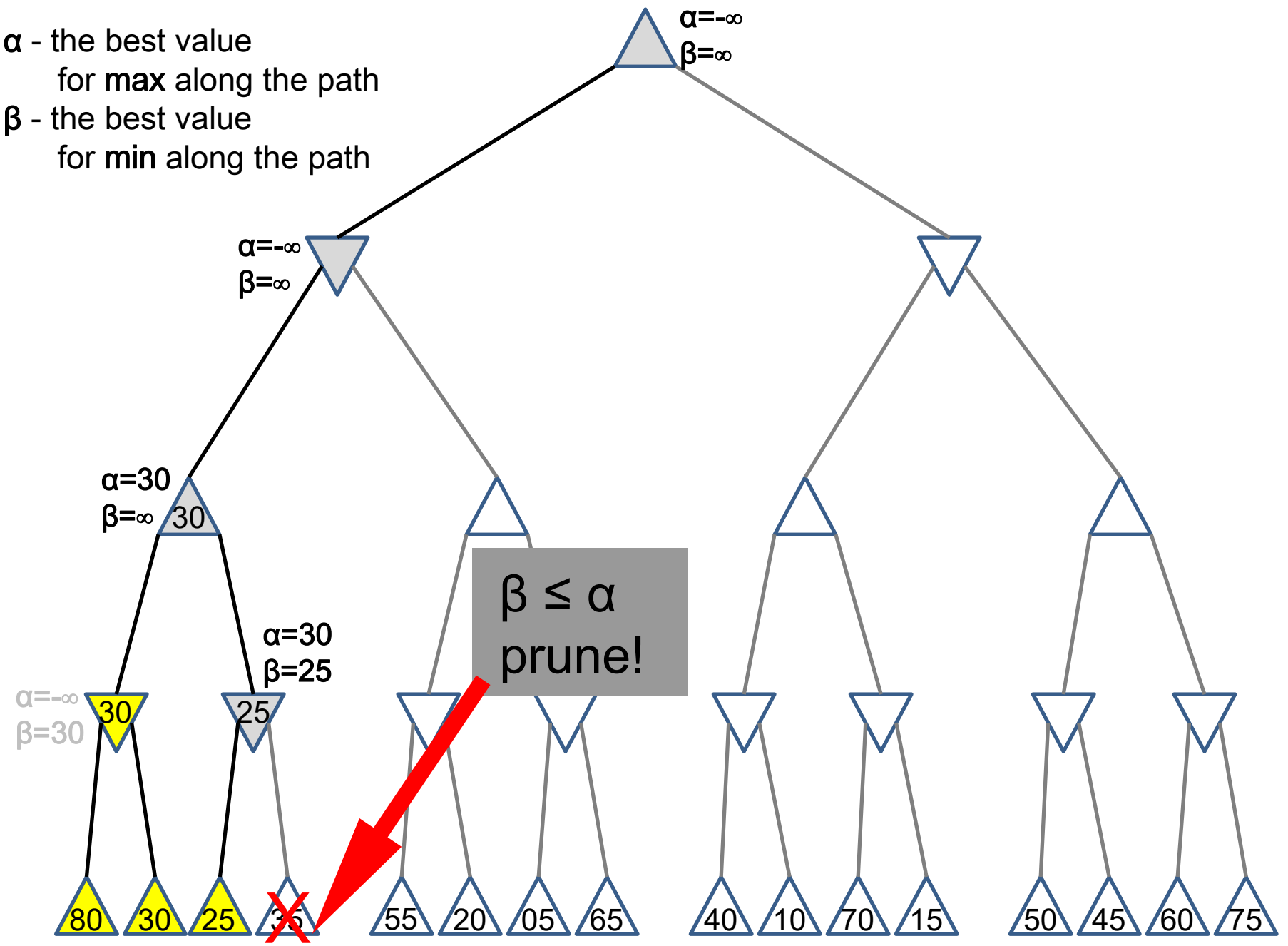


α - the best value
for **max** along the path
 β - the best value
for **min** along the path



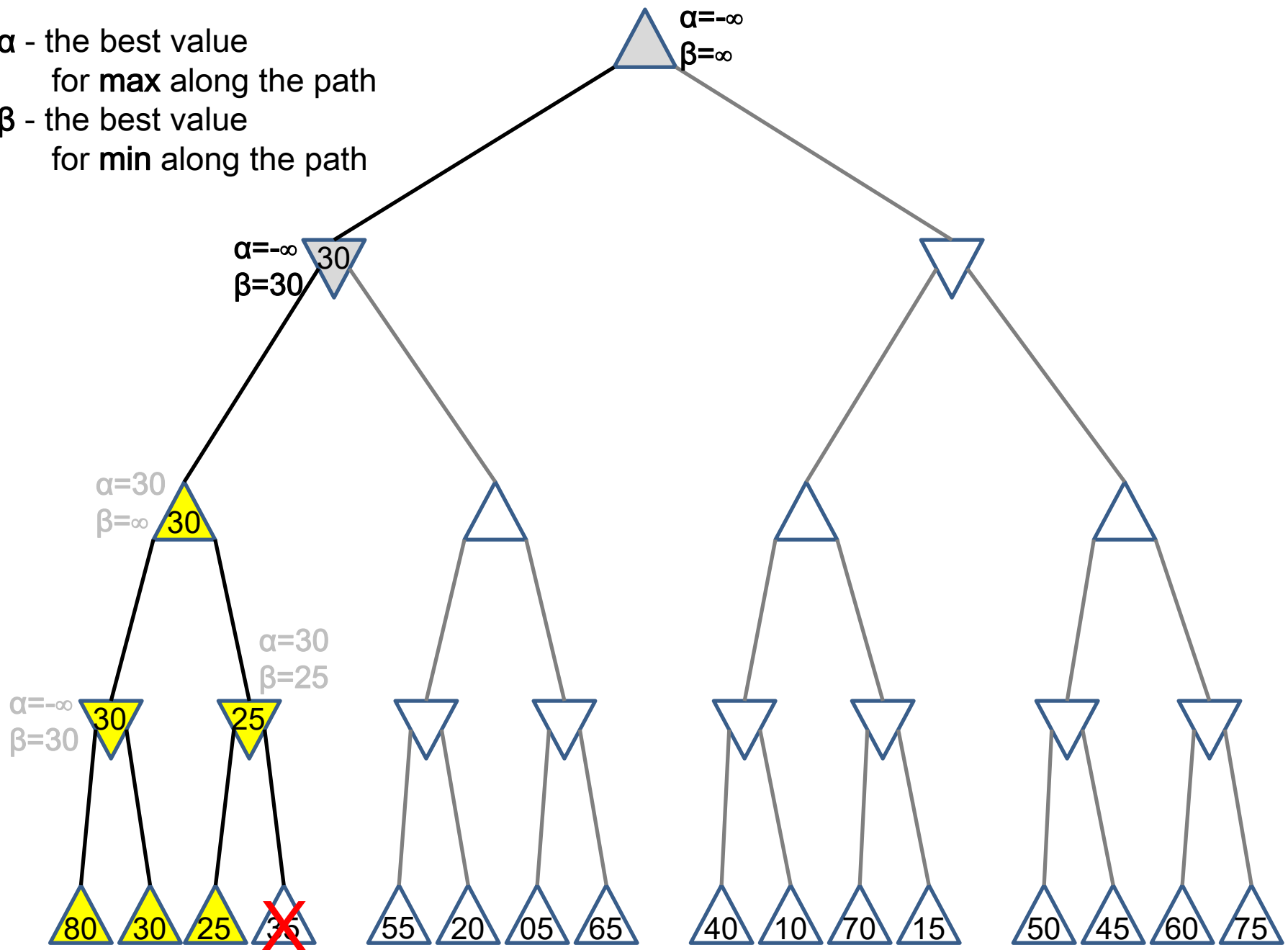
α - the best value
for **max** along the path

β - the best value
for **min** along the path



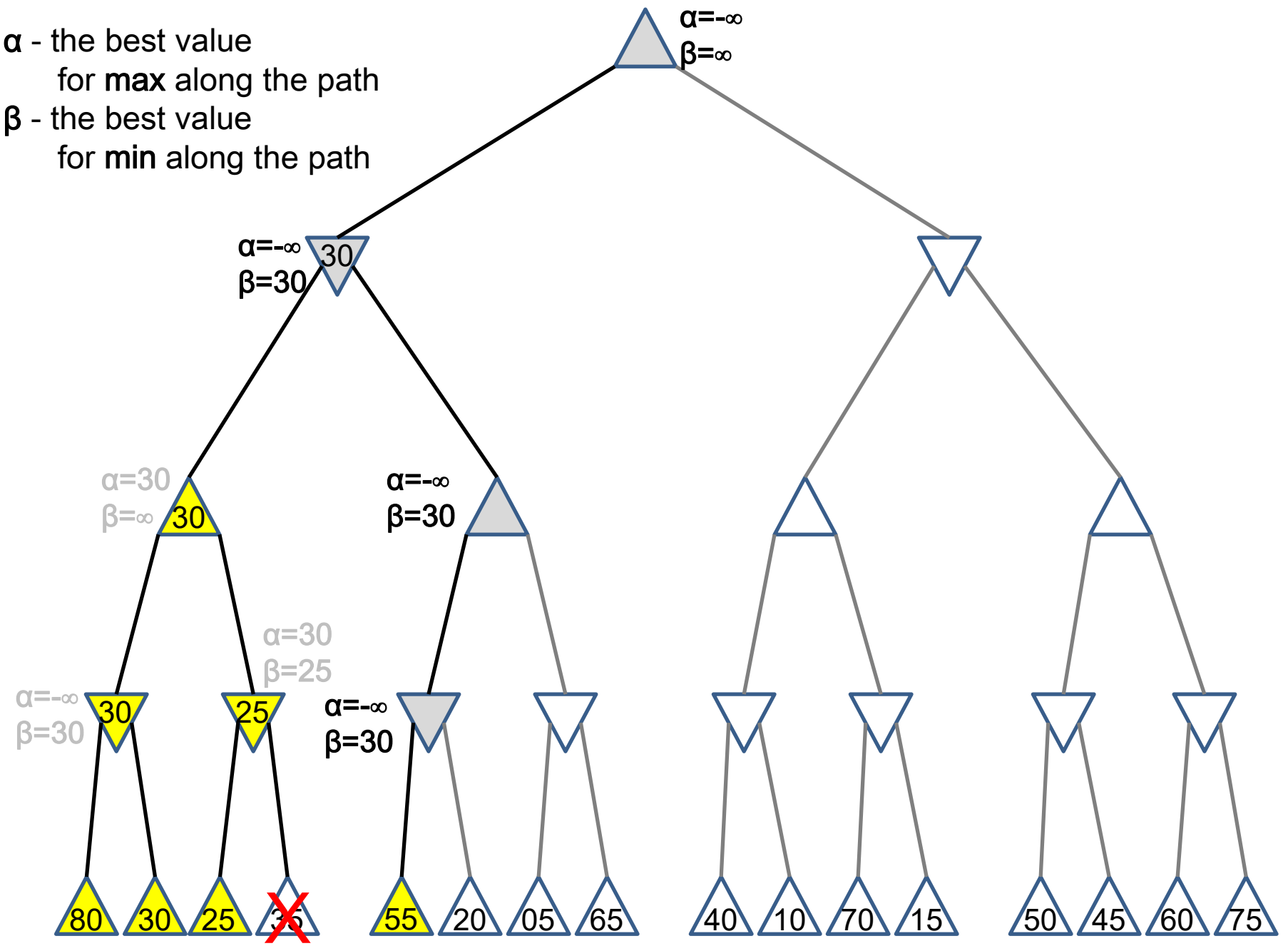
α - the best value
for **max** along the path

β - the best value
for **min** along the path



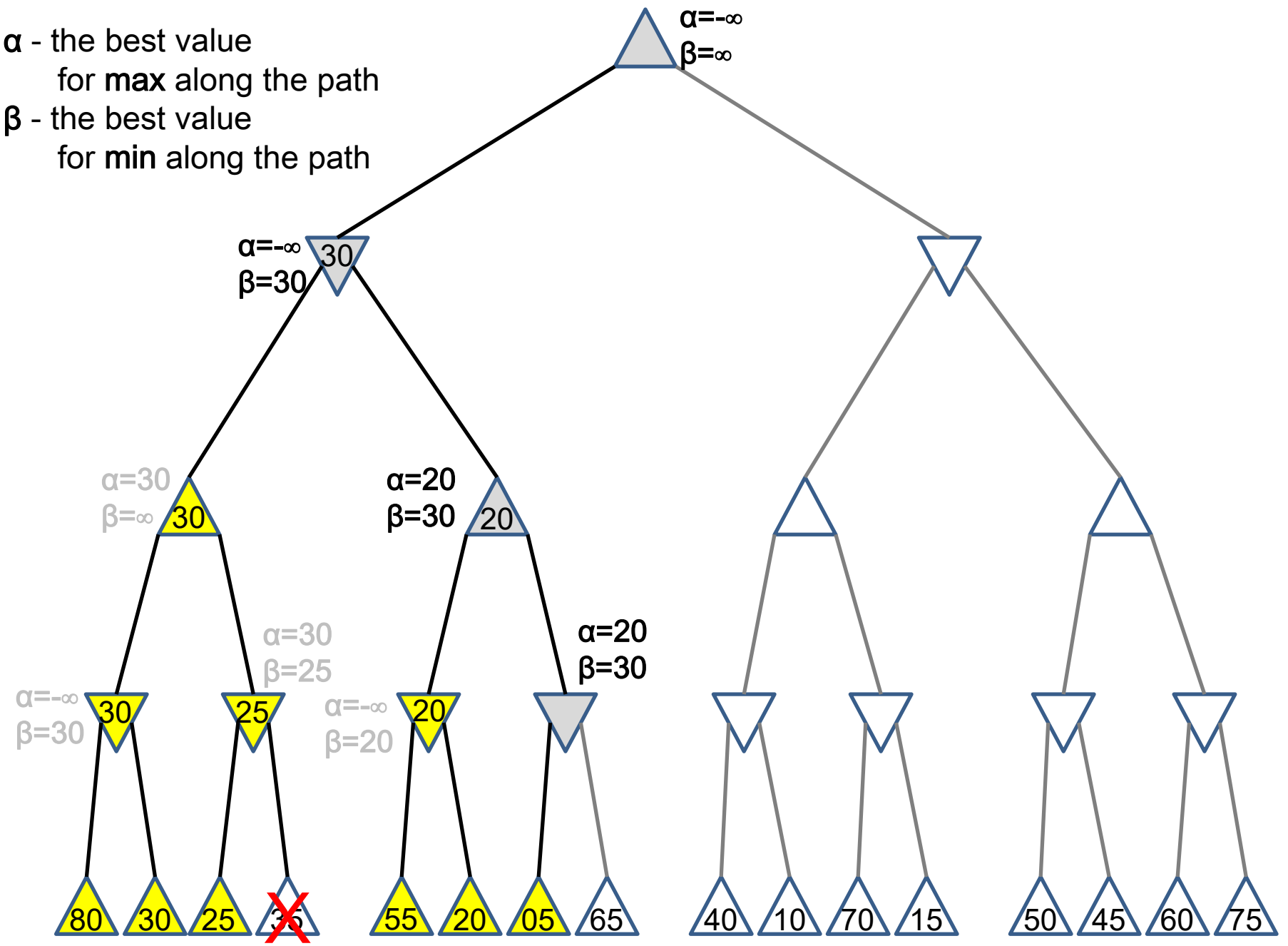
α - the best value
for **max** along the path

β - the best value
for **min** along the path



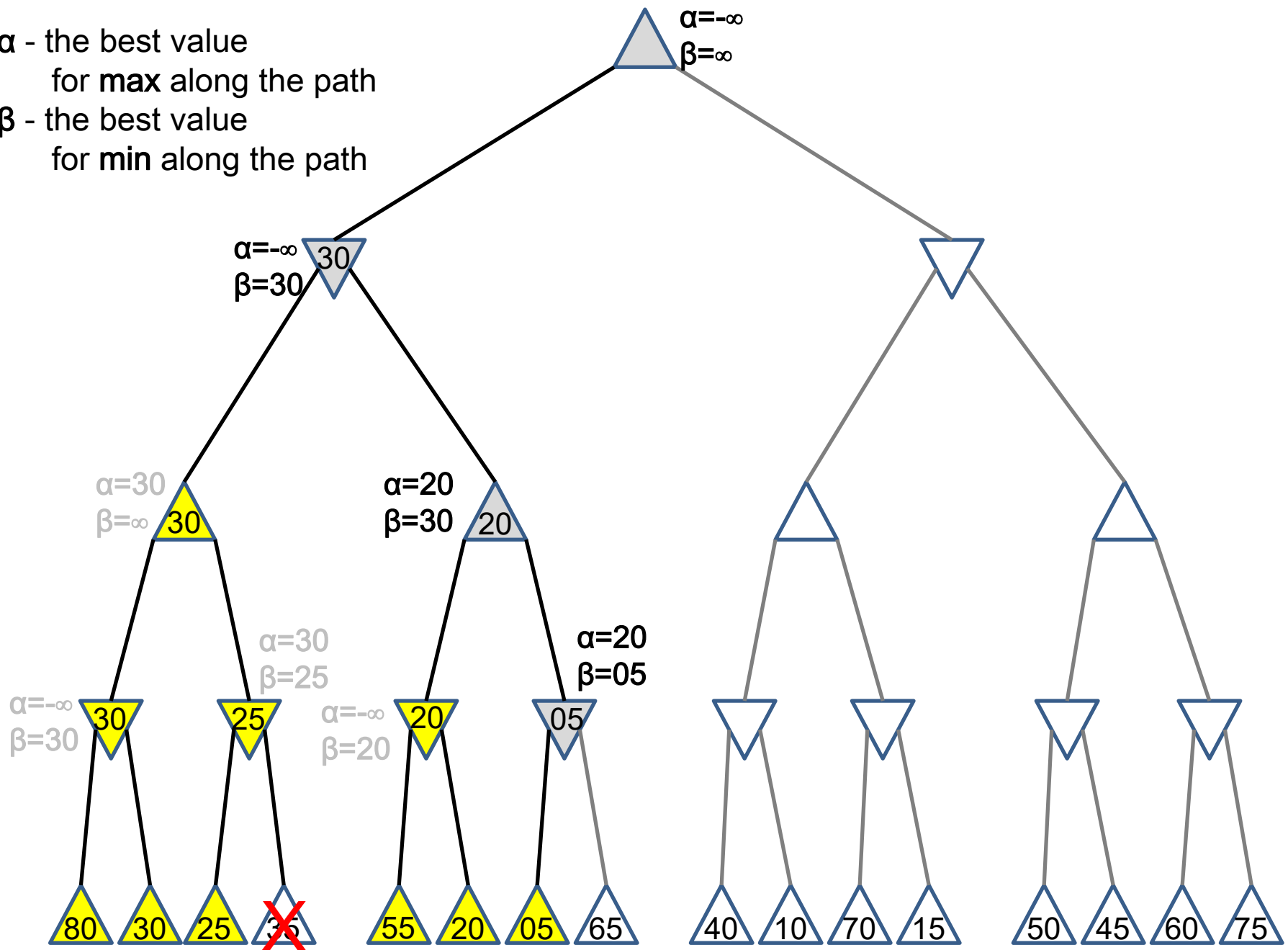
α - the best value
for **max** along the path

β - the best value
for **min** along the path



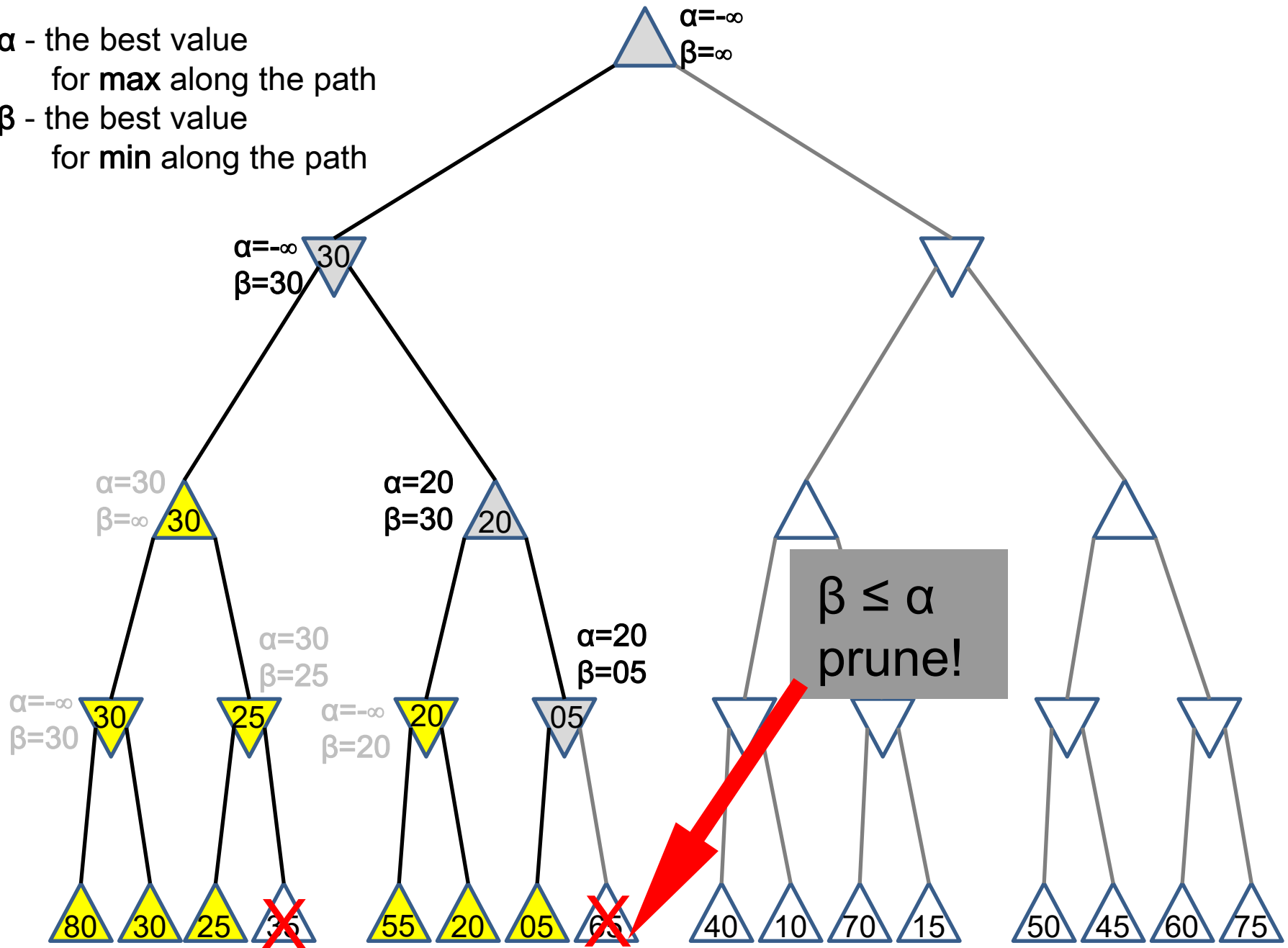
α - the best value
for **max** along the path

β - the best value
for **min** along the path



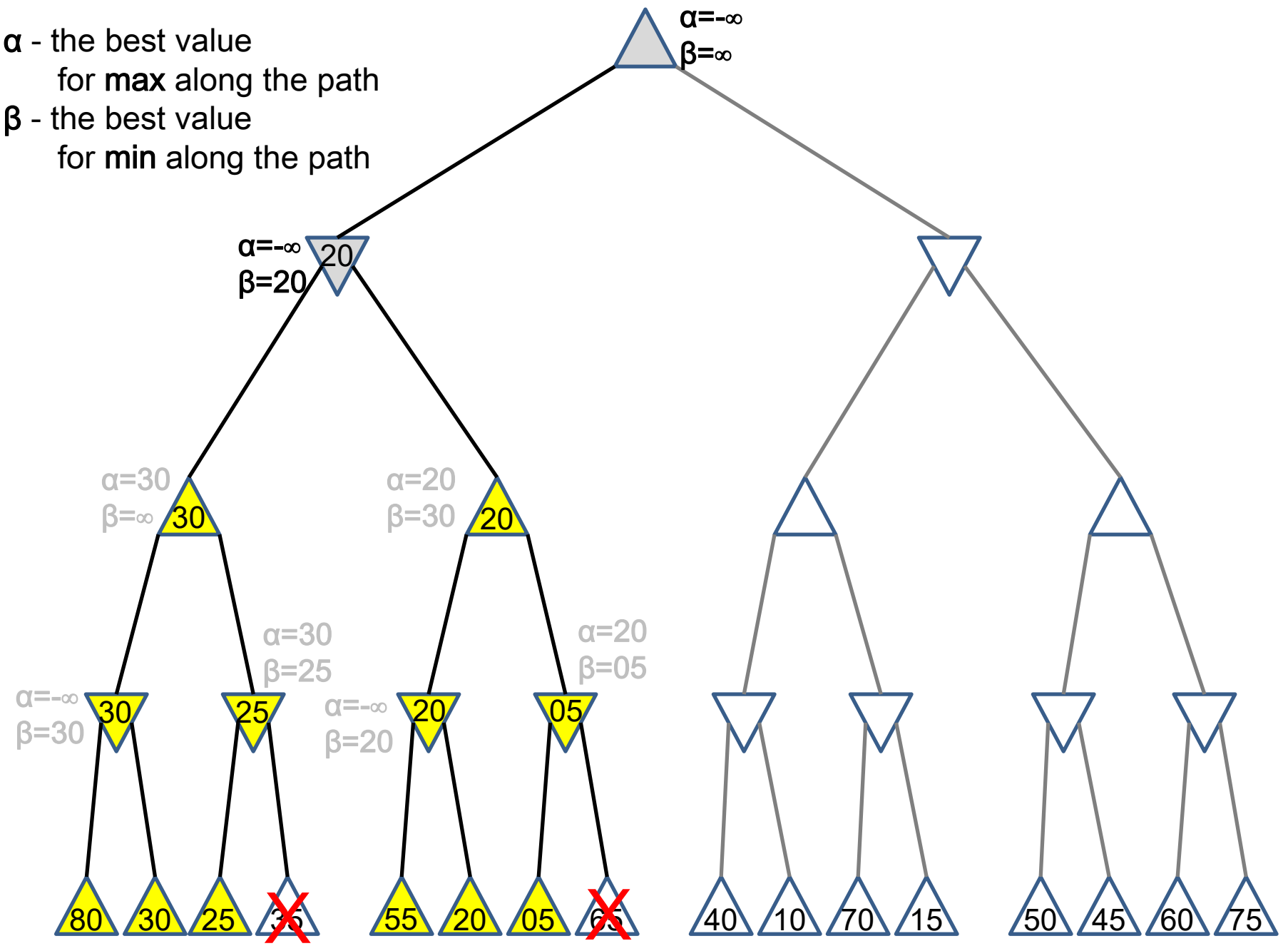
α - the best value
for **max** along the path

β - the best value
for **min** along the path



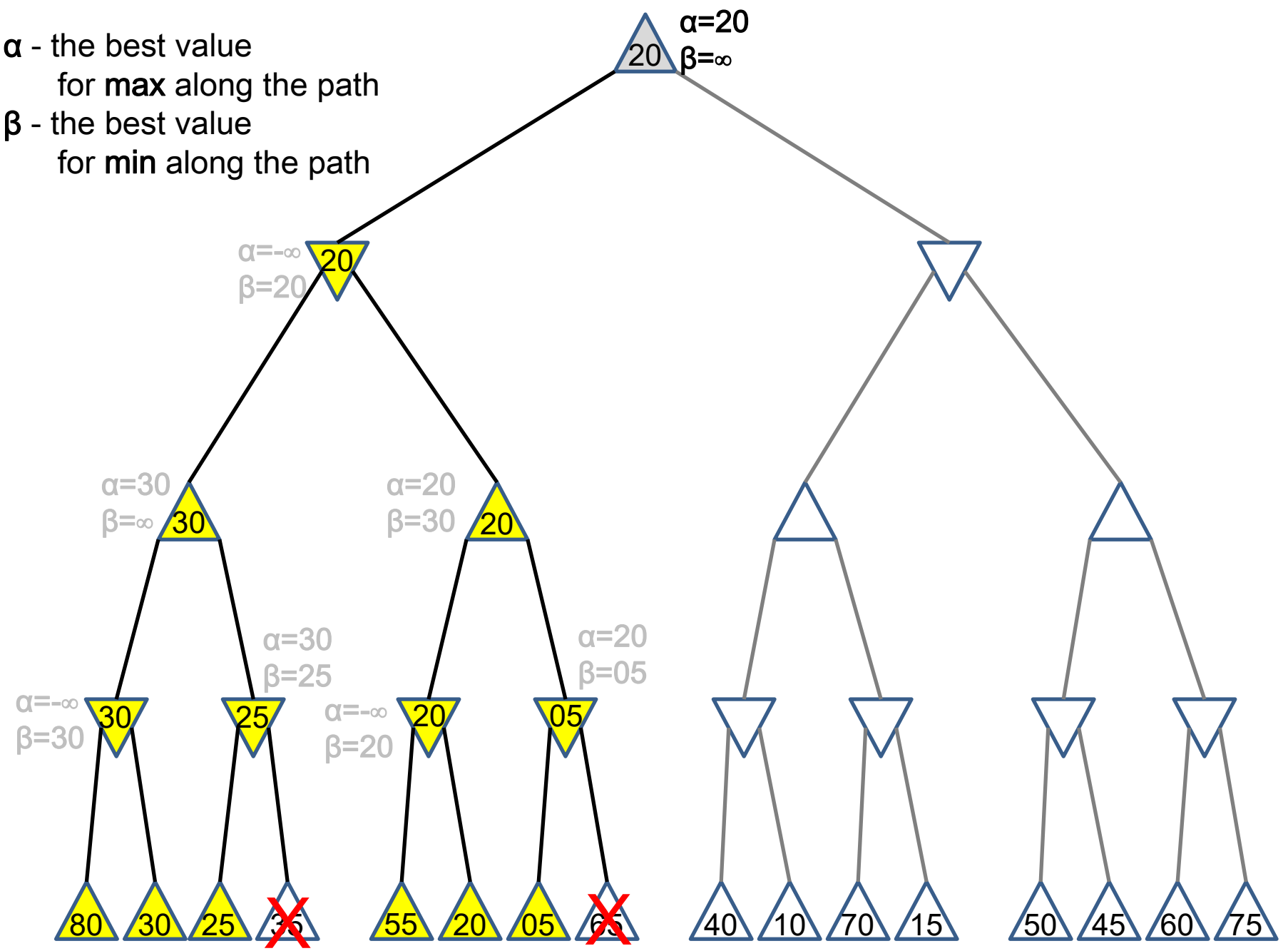
α - the best value
for **max** along the path

β - the best value
for **min** along the path

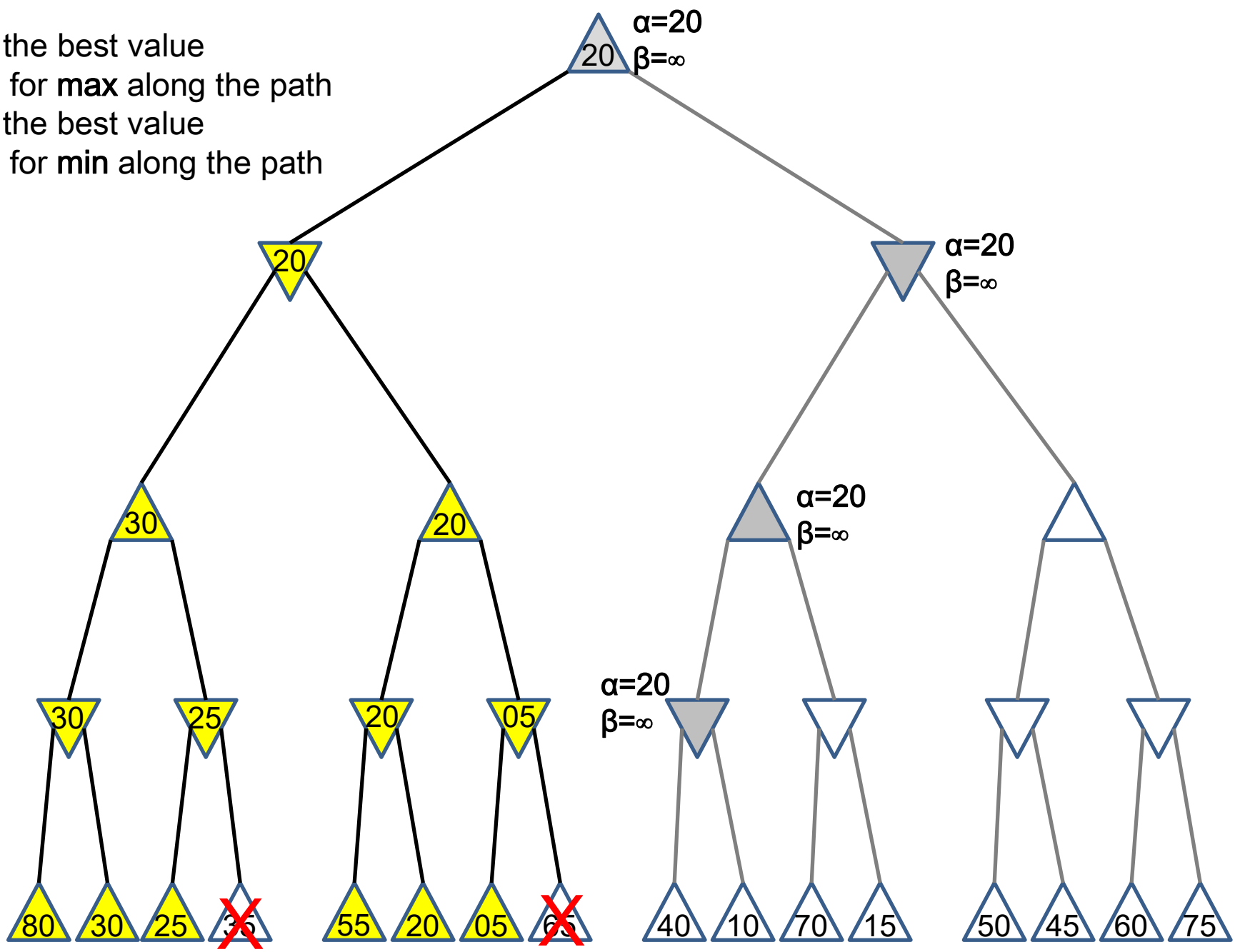


α - the best value
for **max** along the path

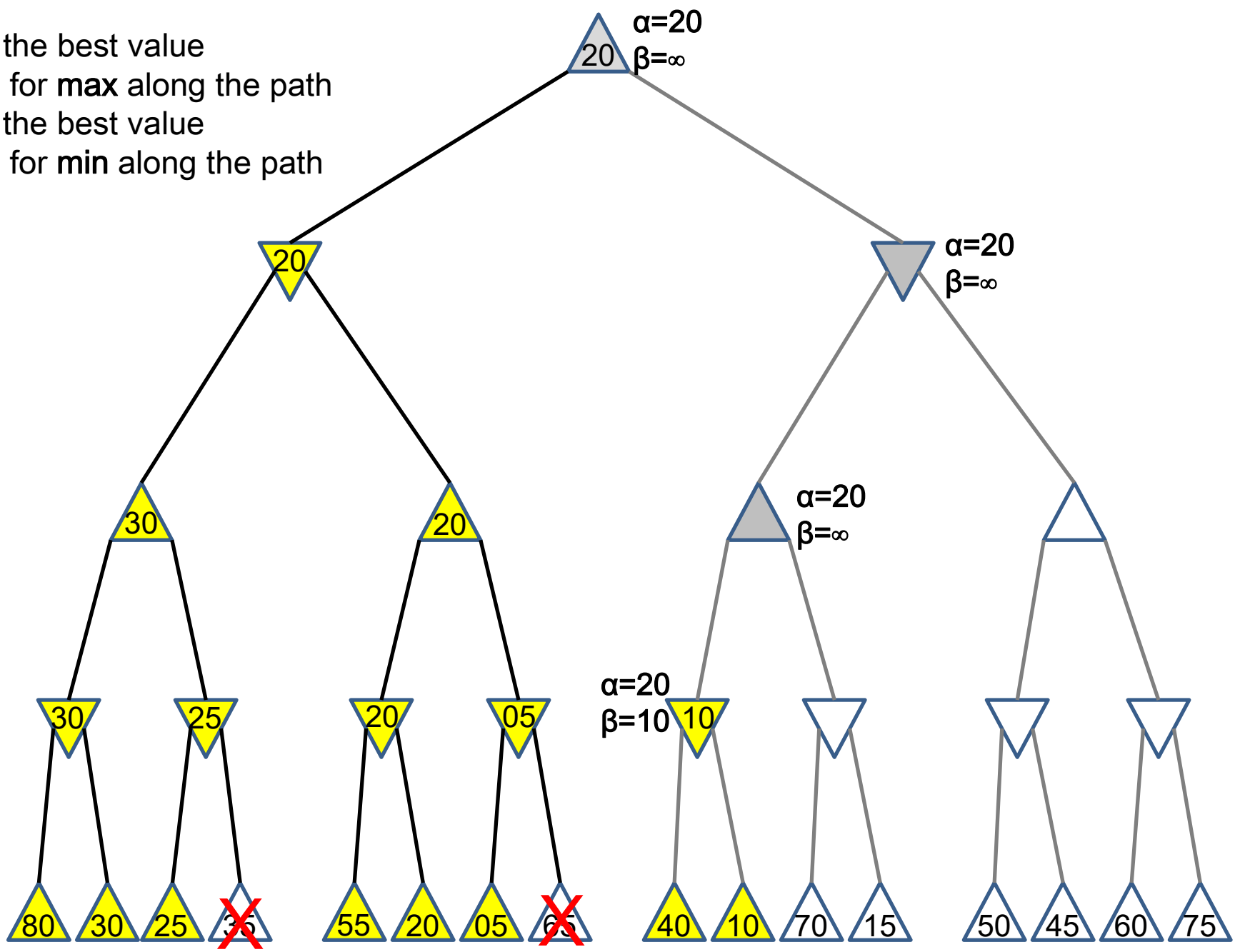
β - the best value
for **min** along the path



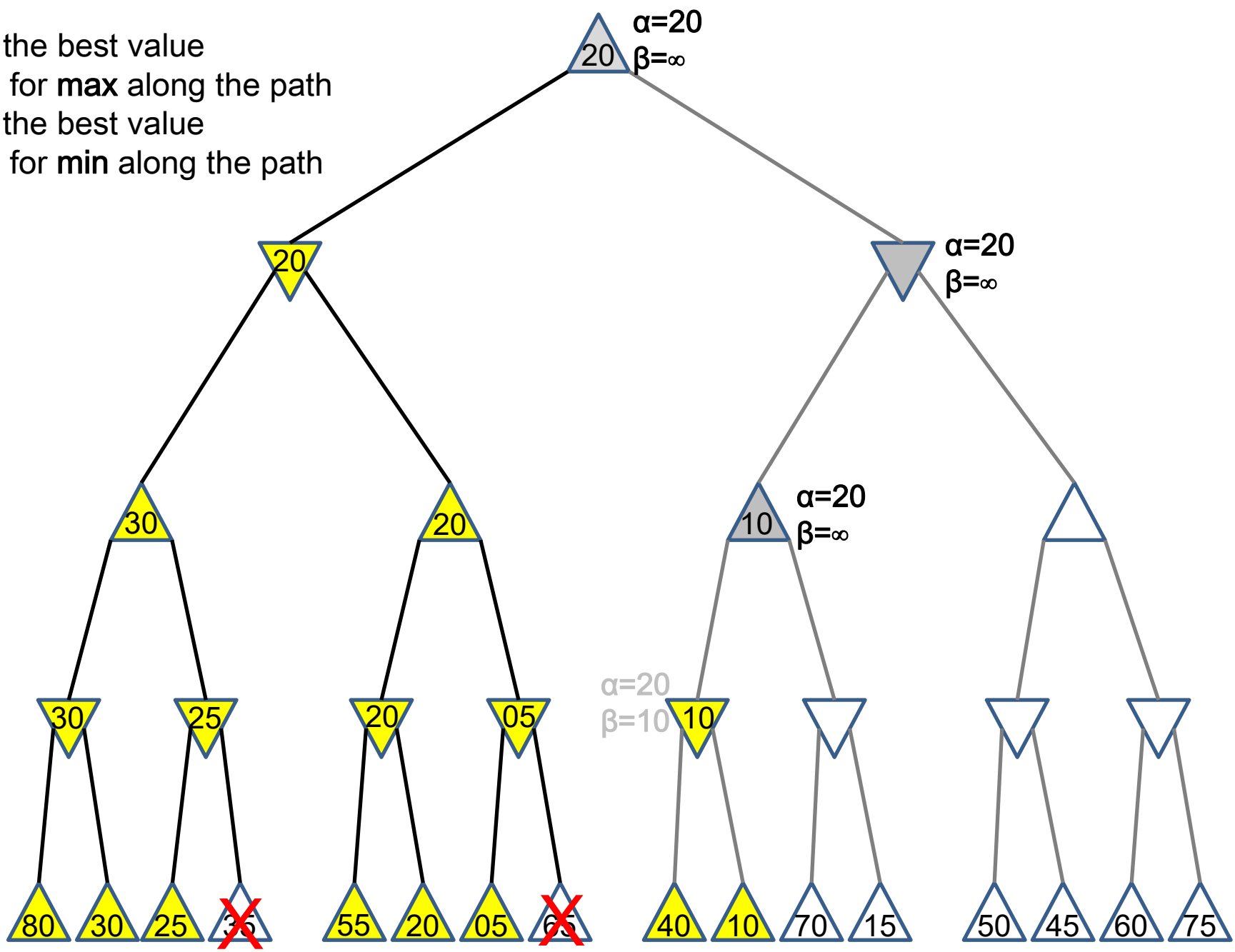
α - the best value
for **max** along the path
 β - the best value
for **min** along the path



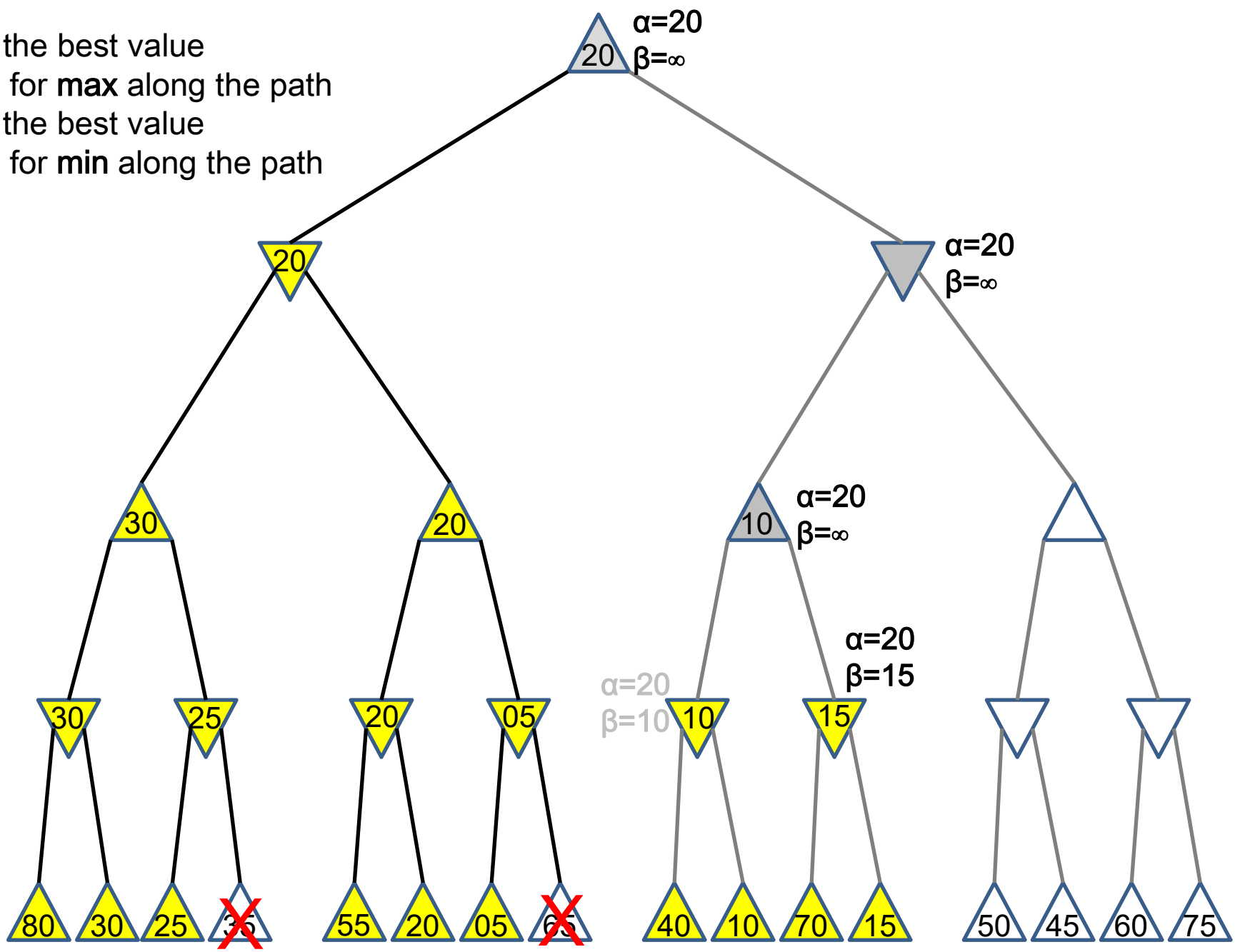
α - the best value
for **max** along the path
 β - the best value
for **min** along the path



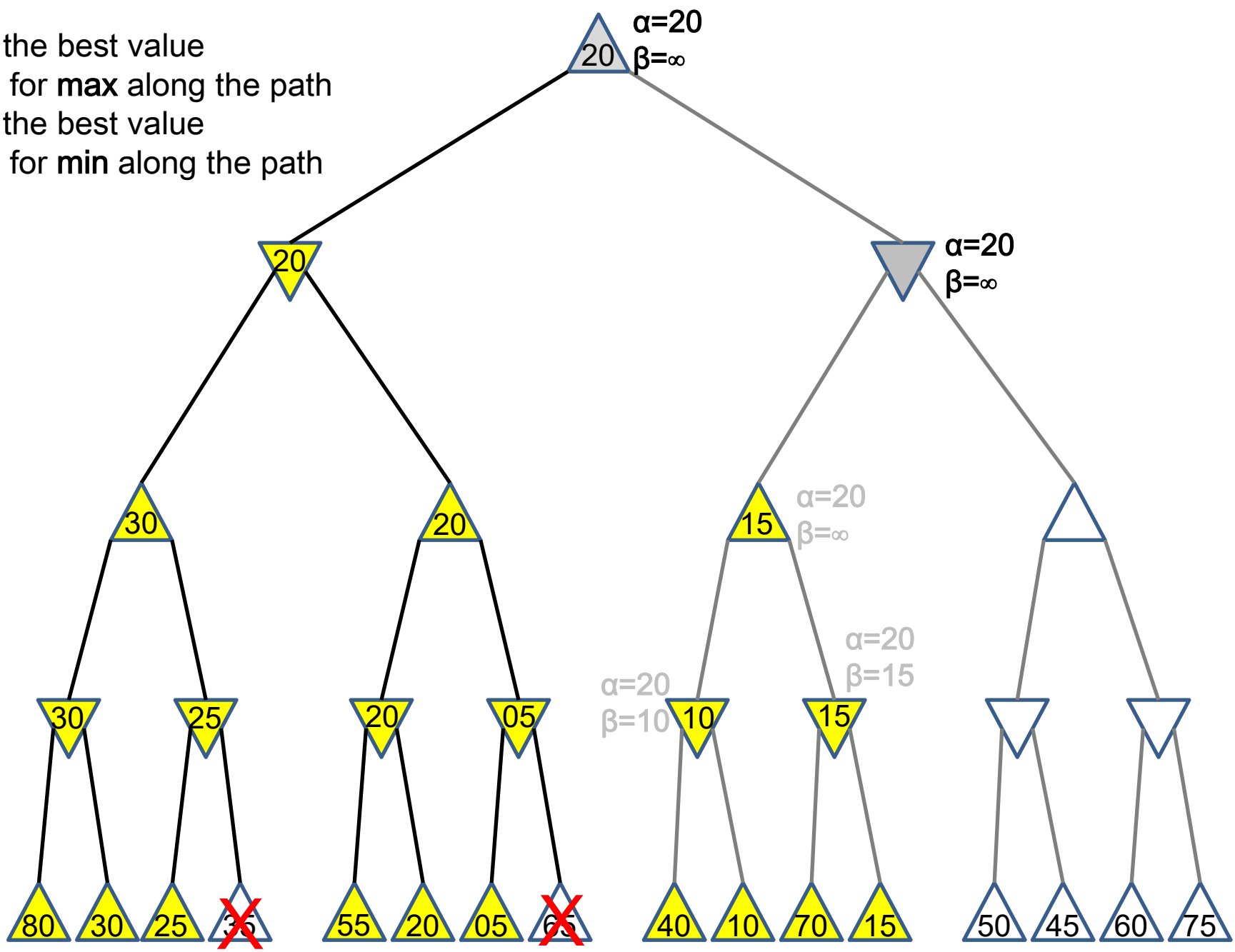
α - the best value
for **max** along the path
 β - the best value
for **min** along the path



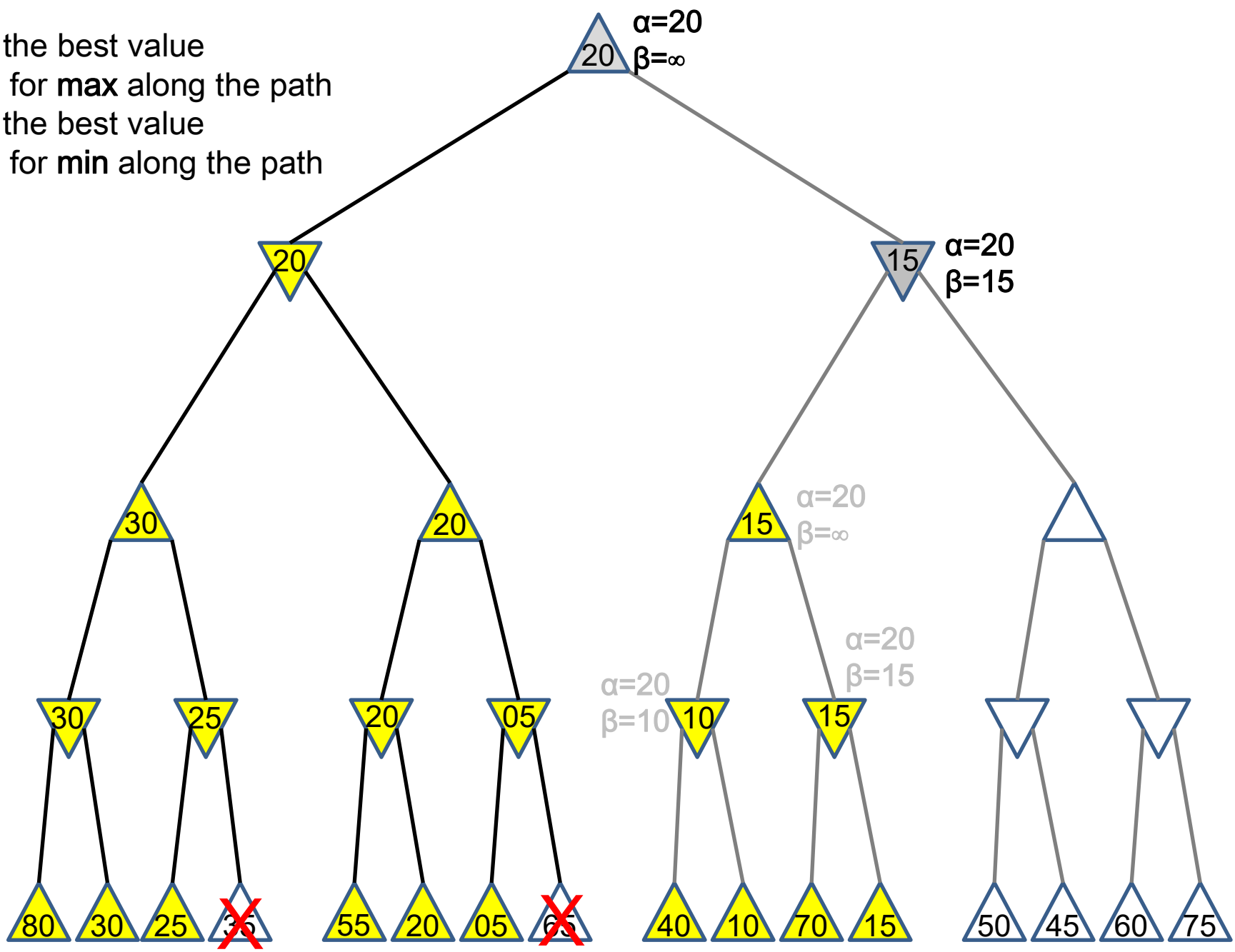
α - the best value
for **max** along the path
 β - the best value
for **min** along the path



α - the best value
for **max** along the path
 β - the best value
for **min** along the path



α - the best value
for **max** along the path
 β - the best value
for **min** along the path



α - the best value
for **max** along the path
 β - the best value
for **min** along the path

